Weather Conditions and Climate Change with ClimateWins - a new Machine Learning Strategy

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Objectives

- Finding new patterns in weather changes over the last 60 years
- Identifying weather patterns outside the regional norm in Europe
- Determining whether unusual weather patterns are increasing
- Generating possibilities for future weather conditions over the next 25 to 50 years based on current trends
- Determining the safest places for people to live in Europe within the next 25 to 50 years

Thought Experiments – Proposal

- Experiment 1 – Detect regional signatures of unusual weather patterns (Deep learning & Clustering)

- Experiment 2 – Project future weather trends (Deep learning-GAN)

- Experiment 3 – Predict the safest regions (Random forest)

Climate signatures

Goal: Identify and quantify deviations from the regional climate norm over time to better understand the emerging signatures of climate anomalies

Specifications: This experiment focuses on detecting and analyzing unusual weather behaviors at a regional level — for example, sudden increases in rainfall or shifts in temperature variability. These signals often precede major climatic disruptions and are key to understanding local climate change impacts. Clustering allows for grouping regions with similar climate dynamics, while deep learning highlights spatial-temporal irregularities.

Suggested methods:
- Hierarchical clustering to group regions based on multivariate weather characteristics (e.g., temperature, pressure, precipitation). Allows the formation of region-specific climate signatures.
- Convolutional Neuronal Network (CNN) to process gridded climate data (like temperature maps) and detect emerging spatial anomalies over time.

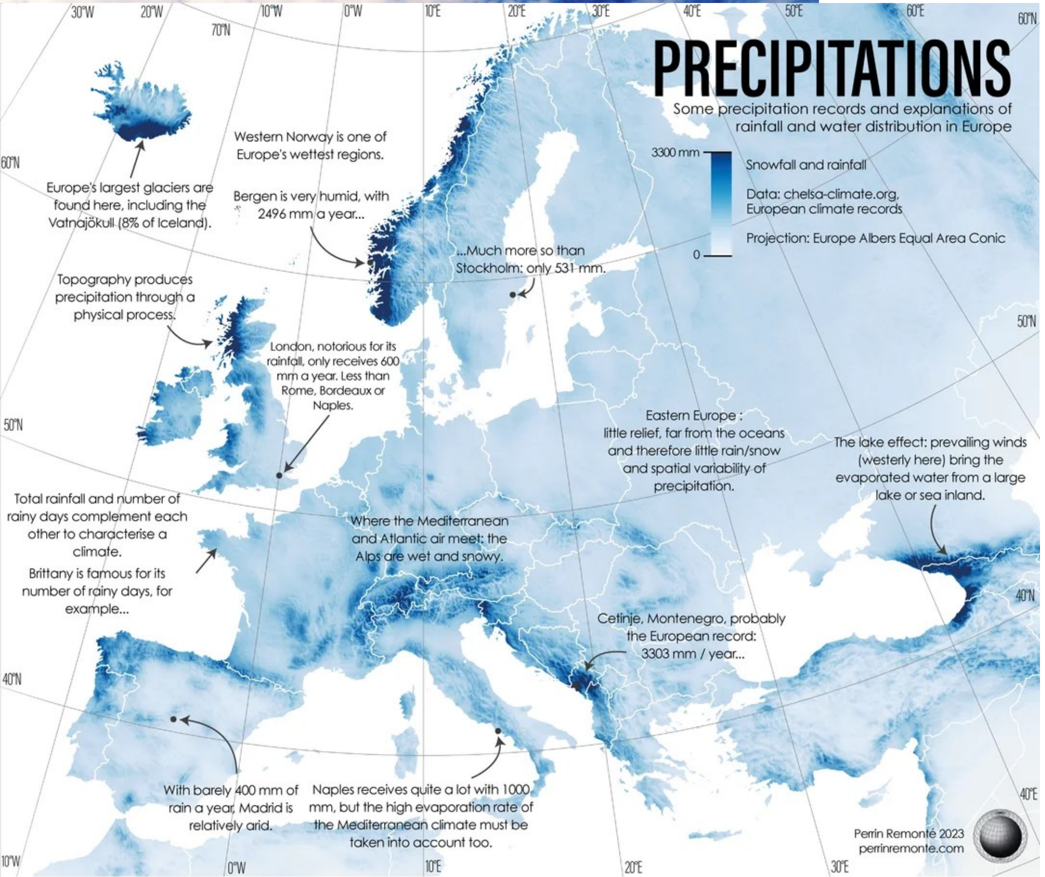
Suggested data: https://cds.climate.copernicus.eu/datasets
ERA5 hourly data, satellite observations

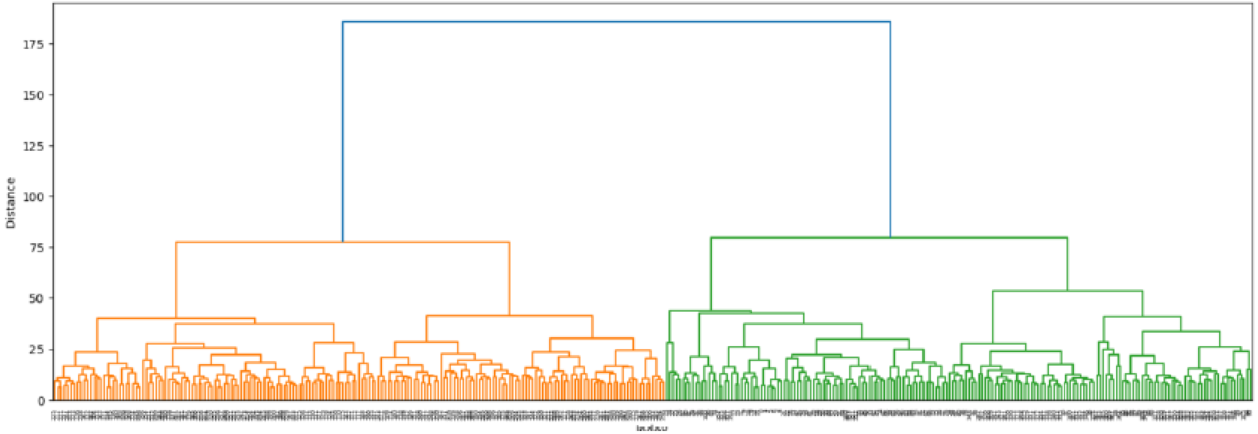
Experiment 1

Machine learning tools

Climate signatures

Hierarchical clustering can segregate regions in Europe based on similar weather records. Some of the subclusters may represent the weather anomalies that emerged recently in the global warming era. Based on the cutoff we set on the dendrogram, we can explore different depth of the data regularities.





CNNs can recognize visual patterns like those from satellite pictures, temperature, precipitation, etc value-maps. So, if there is unexpected pattern in e.g. rainfall or drought ditribution it would be picked up by the model. One can feed in the weather radars data like the heatmap on the left.

Experiment 2
Overview

Future climate

Goal: Generate realistic, future-oriented weather trend projections based on learned spatiotemporal climate patterns

Specifications: Climate modelling is computationally expensive and sometimes limited in resolution or adaptability. This experiment uses advanced generative models to learn complex time-based patterns from past data and produce high-resolution forecasts under various scenarios

Suggested methods:
- Recurrent Neural Networks (RNNs) to model sequences of weather observations over time. LSTM variants can capture long-term dependencies
- Generative Adversarial Networks (GANs) combined with RNNs to generate realistic future weather data by learning from existing time series (GAN-RNN hybrid). Enables data-driven simulation of possible future scenarios.

Suggested data: https://cds.climate.copernicus.eu/datasets
ERA5 hourly data, satellite-derived temperature, wind and precipitation fields

Experiment 2

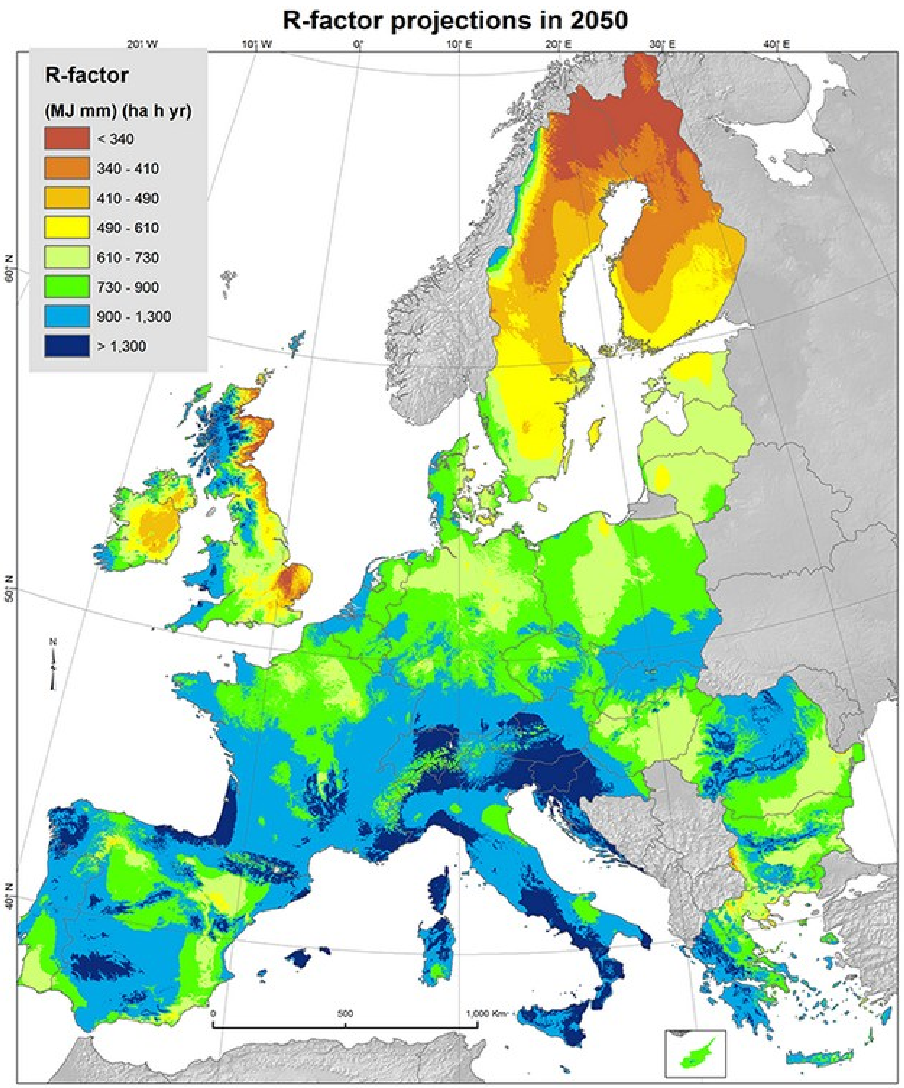
Machine learning tools

Future climate

RNNs can process time series data and they could infer the future weather conditions based on the past. This model can help establishing temporal dependencies within a variable and across many variables.

GANs can produce realistic synthetic data which can be then incorporated into the broader weather modelling frameworks to enhance forecasting, fill data gaps, or simulate future scenarios.

Combining temporal memory with adversarial learning will let the model learn both - when and how weather patterns evolve.



Future climate

GANs trained on weather conditions pictures reached fairly high model accuracy

Experiment 3 Overview

Safest regions

Goal: Predict regions that are least vulnerable to extreme weather events and climate volatility, supporting decision-making for insurance, infrastructure and migration planning

Specifications: As climate risks increase, identifying relatively stable regions becomes crucial. “Safe” could be defined by low frequency of severe weather, moderate temperature shifts, or consistent precipitation patterns. This experiment maps vulnerability levels by region, combining various features and indicators.

Suggested methods:

- Random Forest is a robust ensemble method suitable for multi-feature classification tasks. Can rank importance of different risk factors and output interpretable results.

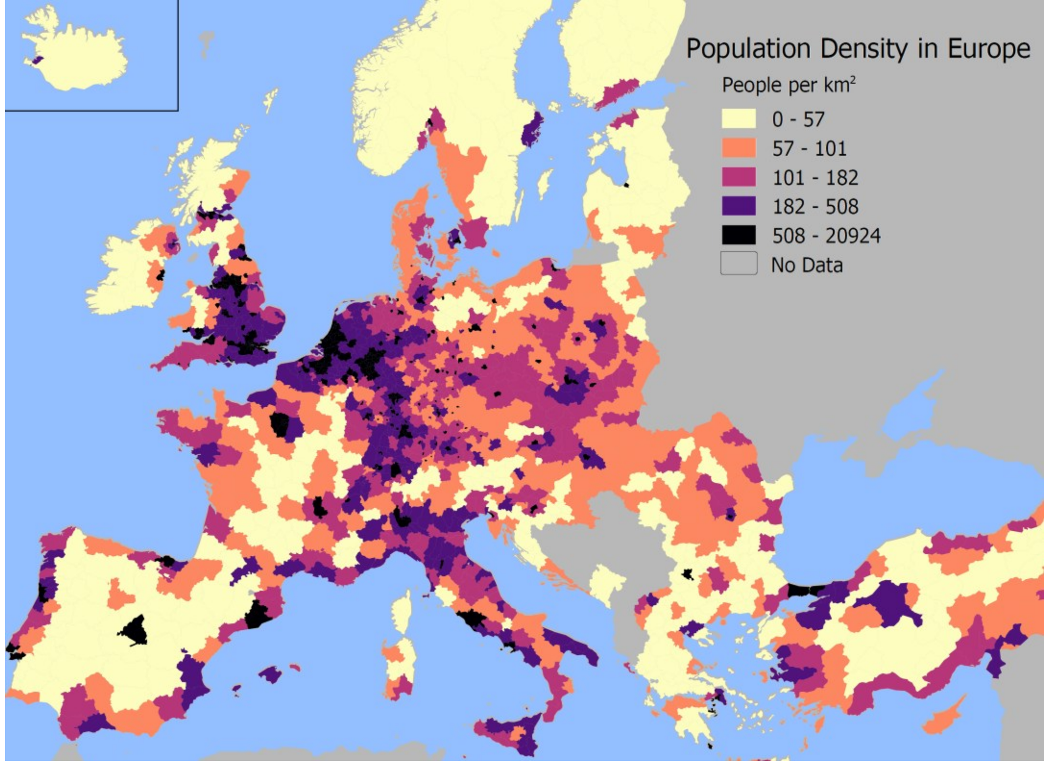
Suggested data: https://cds.climate.copernicus.eu/datasets
ERA5 hourly data, population density, infrastructure data

Experiment 3

Machine learning tools

Safest regions

Random Forest is a machine learning algorithm made up of many decision trees. By asking a series of climate and safety relevant questions the model can reveal where certain unfavourable conditions meet.



The decion trees may include features like:

- Population density
- Precipitation
- Wind speed
- Temperature change
- Elevation
- Distance from the coast

… and many more!

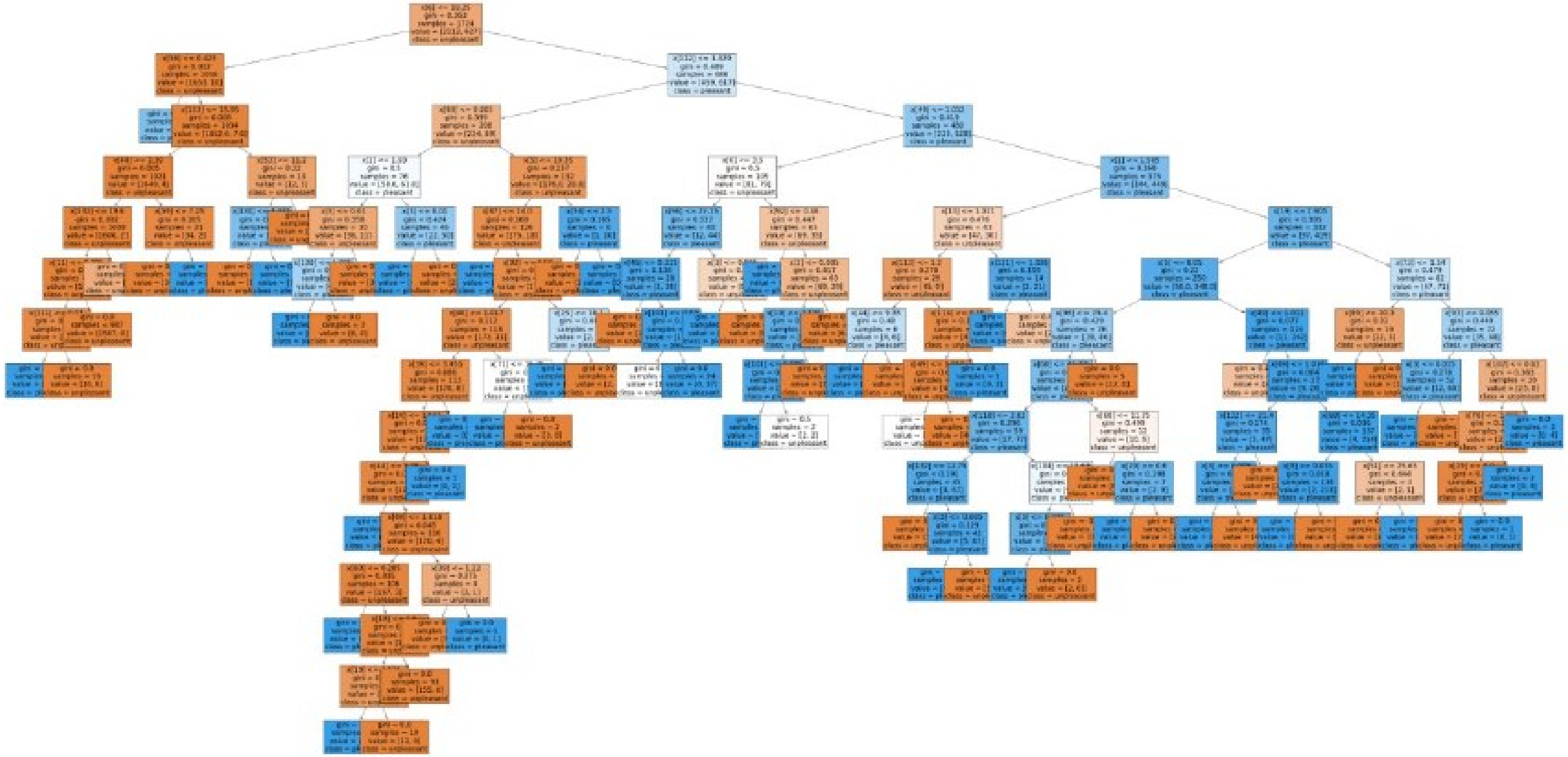
Experiment 3

Machine learning tools

Future climate

Random forest applied on the weather data gave very good results. The model classified the pleasent/unpleasent weather from 15 weather stations based on many features.

Accuracy 96%



Summary & Recommendations

- ClimateWins will benefit from implementing the new ML strategies into their portfolio – this innovative approach provides accurate, fast and easy to implement methods to model the weather trends in Europe and project them decades ahead

- The most promising algorithms tested on the weather data were:
  - Hierarchical clustering
  - GANs
  - Random forest

- Reccomendation for ClimateWins is to apply:
  - Hierarchical clustering to capture the local weather anomalies
  - GANs for modelling future weather conditions
  - Random forest to find safest areas to live in the future

- Next step will be to implement additional data from:
  - other weather databases
  - satellites
  - socio-economical maps

THANK YOU!

If you have any questions, please contact me:

The GitHub repository with scripts and data for methods used in this project can be found here: